Sprague–Grundy 函数论文

标准

- **完整 SG 分析**：在相应定理明确覆盖的参数范围或游戏子类中，能够确定每个合法局面的 SG 值。
- 因此，一篇论文不必解决其所定义的全部推广游戏；只要它对一个边界清楚的参数族给出任意合法局面的完整 SG 计算方法，就纳入本表。

论文对比

年份	BibTeX 键	论文	分析的具体游戏	主要 SG 结果	适用范围与局限
2025	`NivR2025`	Nivasch and Rubinstein, *A Convergence Technique for the Game (i)-MARK*	(i)-Mark(({1}, {d_1,d_2}))	提出基于收敛技术的多项式时间算法，可以计算任意位置 (n) 的 SG 值，从而在算法意义上完整确定该子类中每个游戏的 SG 函数。	该结果是一种高效的计算刻画，而不是简单的闭式公式；它也不能覆盖 (S) 和 (D) 的任意取值。
2021	`BorGHM2021`	Boros et al., *On the Sprague–Grundy Function of Extensions of Proper Nim*	Proper Nim 的选择性复合扩展，也称 Tetris 型扩展	对至少包含三个堆的相应复合游戏给出完整的 SG 闭式公式。	公式仅适用于 (n\geq3)；(n=2) 的情形更加复杂，论文只给出了部分结果。
2019	`BorGHMM2019a`	Boros et al., *Sprague–Grundy Function of Symmetric Hypergraphs*	对称超图 Nim（symmetric hypergraph Nim）	刻画其 SG 函数满足 Jenkyns–Mayberry 公式的对称超图；对于这些游戏，该公式能够确定每个局面的 SG 值。	结果对所刻画的 JM 类是完整的，但没有为所有对称超图给出统一的 SG 公式。
2019	`BorGHMM2019b`	Boros et al., *Sprague–Grundy Function of Matroids and Related Hypergraphs*	拟阵及相关超图上的 Nim（matroid and related hypergraph Nim）	为若干超图类别给出显式 SG 公式，并刻画该公式适用的图和拟阵。	公式完整解决了满足相应条件的类别，包括论文所述的自对偶拟阵情形，但不能处理任意超图 Nim。

年份	BibTeX 键	论文	分析的具体游戏	主要 SG 结果	适用范围与局限
2018	`BorGHMM2018`	Boros et al., *On the Sprague–Grundy Function of Exact (k)-Nim*	Exact (k)-Nim，即每一步必须恰好减少 (k) 个堆	当 (2k\geq n) 时，给出每个局面 SG 值的显式公式及其高效计算方法。	对于剩余的 (2k<n) 参数范围，论文没有给出一般性的完整刻画。
2016	`Sop2016`	Sopena, *(i)-Mark: A New Subtraction Division Game*	整数型减法—整除游戏 (i)-Mark((S,D))：从 (n) 可移动到 (n-s)，或在 (d\mid n) 时移动到 (n/d)	对 (i)-Mark(([1,t-1],{t}))、(i)-Mark(([1,t-1],{2})) 以及若干 (i)-Mark(({a,2a},{2})) 情形刻画 SG 序列的近似周期结构，并由此在 (O(\log n)) 时间内计算相应定义域中任意局面 (n) 的 SG 值。	完整计算结论只适用于论文明确证明的参数族；论文证明一般有限 (S,D) 的 SG 序列非周期，但没有解决任意 (S,D) 的完整 SG 函数。
2011	`Fra2011`	Fraenkel, *Aperiodic Subtraction Games*	Mark：从整数 (n) 可移动到 (n-1) 或 (\lfloor n/2\rfloor)	借助整数二进制表示刻画 Mark 的三值 SG 序列，并给出计算任意 (n) 的 SG 值的线性时间算法，因此也能处理多个 Mark 局面的和。	完整 SG 算法针对 Mark；论文对 UpMark 和 Mark-(t) 的讨论主要给出必胜策略与非周期性，不能统一视为这些推广游戏的完整 SG 解。
2011	`cairns2011`	Cairns, Ho, and Lengyel, *The Sprague–Grundy Function of the Real Game Euclid*	原始 Euclid game：局面为一对非负整数，每步从较大数中减去较小数的正整数倍，并以某一分量变为零作为终止条件	给出原始 Euclid game 的显式 SG 公式，可以确定定义域内任意整数对的 SG 值；同时说明该函数与“两个分量相等时终止”的 Euclid 变体之间的关系。	公式依赖原始 Euclid game 的移动规则与终止条件，不能直接用于其他 Euclid 变体。

Table 1

BibTeX 条目

```bibtex
@article{NivR2025,
    title     = {{A Convergence Technique for the Game {$i$}-{MARK}}},
    author    = {Gabriel Nivasch and Oz Rubinstein},
    journal   = {Theoretical Computer Science},
    volume    = {1057},
    pages     = {115557},
    year      = {2025}
}

@article{BorGHM2021,
    title     = {{On the Sprague--Grundy Function of Extensions of Proper {Nim}}},
    author    = {Endre Boros and Vladimir Gurvich and Nhan Bao Ho and Kazuhisa Makino},
    journal   = {International Journal of Game Theory},
    volume    = {50},
    number    = {3},
    pages     = {635--654},
    year      = {2021}
}

@article{BorGHMM2019a,
    title     = {{Sprague--Grundy Function of Symmetric Hypergraphs}},
    author    = {Endre Boros and Vladimir Gurvich and Nhan Bao Ho and Kazuhisa Makino and Peter Mursic},
    journal   = {Journal of Combinatorial Theory, Series A},
    volume    = {165},
    pages     = {176--186},
    year      = {2019}
}

@article{BorGHMM2019b,
    title     = {{Sprague--Grundy Function of Matroids and Related Hypergraphs}},
    author    = {Endre Boros and Vladimir Gurvich and Nhan Bao Ho and Kazuhisa Makino and Peter Mursic},
    journal   = {Theoretical Computer Science},
    volume    = {799},
    pages     = {40--58},
    year      = {2019}
}

@article{BorGHMM2018,
    title     = {{On the Sprague--Grundy Function of Exact {$k$}-{Nim}}},
    author    = {Endre Boros and Vladimir Gurvich and Nhan Bao Ho and Kazuhisa Makino and Peter Mursic},
    journal   = {Discrete Applied Mathematics},
    volume    = {239},
    pages     = {1--14},
    year      = {2018}
}

@article{Sop2016,
```

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    title     = {{$i$-Mark: A new subtraction division game}},
    author    = {Eric Sopena},
    journal   = {Theoretical Computer Science},
    volume    = {627},
    pages     = {90--101},
    year      = {2016}
}

@article{Fra2011,
    title     = {{Aperiodic subtraction games}},
    author    = {Aviezri S. Fraenkel},
    journal   = {Electronic Journal of Combinatorics},
    volume    = {18},
    number    = {19},
    pages     = {1--12},
    year      = {2011}
}

@article{cairns2011,
    title     = {{The Sprague--Grundy function of the real game Euclid}},
    author    = {Grant Cairns and Nhan Bao Ho and Tam{\'a}s Lengyel},
    journal   = {Discrete Mathematics},
    volume    = {311},
    number    = {6},
    pages     = {457--462},
    year      = {2011},
    publisher = {Elsevier}
}
```

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Complete Sprague–Grundy functions are known for several restricted families of impartial games. For the Mark game, Fraenkel characterized its ternary SG sequence through the binary representation of the heap size and obtained a linear-time algorithm for computing the SG value of any position \cite{Fra2011}.
Sopena subsequently introduced the integral variant (i)-Mark and derived a SG algorithms for several parameterized subfamilies whose SG sequences are almost periodic \cite{Sop2016}.
For (i)-Mark(({1},{d_1,d_2})), Nivasch and Rubinstein later gave a convergence-based algorithm for computing the SG value of any position \cite{NivR2025}.
Within the Nim family, Boros et al. obtained an SG formula for proper Nim with at least three piles \cite{BorGHM2021}.
Their related work characterized symmetric hypergraph Nim instances \cite{BorGHMM2019a} and several matroid and related hypergraph Nim classes \cite{BorGHMM2019b} governed by the Jenkyns–Mayberry formula.
The same authors also derived an explicit SG formula for Exact (k)-Nim when (2k\geq n) \cite{BorGHMM2018}.
For the Euclid game, Cairns, Ho, and Lengyel established an explicit SG formula for every position and related it to the SG function of the variant \cite{cairns2011}.
These results determine the SG value of any position within some game classes or parameter regimes, although none extends to all Mark, hypergraph Nim, Exact (k)-Nim, or Euclid variants.